लोक सेवा आयोग,
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उपसमूहको प्रतियोगितात्मक लिखित परीक्षा
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1. **What are the various systems of classification of roads? Briefly outline the classifications based on location and function. Discuss in details the provision made in Nepal Road Standards (NRS) with respect to functional classification of Road Network of Nepal.**

Answer:

According to NRS 2070, roads in Nepal are broadly classified into administrative classification and technical classification.

i. Administrative classification:

- ➢ It is important for assigning national importance and level of government responsible for overall management and methods of financing.
- ➢ According to this classification, roads are classified into:

a. National highway:

- ✓ They are the main roads connecting East to west and North to South of the nation.
- ✓ These serve for long distance travel, provide consistently higher level of service in terms of speed, travel and inter community mobility.
- ✓ They are designate by ‘NH’ followed by two-digit number.

b. Feeder roads:

- ✓ These roads serve by connecting district headquarters, major economic centers, tourism centers to national highways or other roads.
- ✓ They are designated by letter F followed by 3-digit number.

c. District roads:

- ✓ These are important roads within a district, serving areas of production and market and connecting with main highways.

d. Urban roads:

- ✓ These roads serve within urban municipalities.
- ✓ According to National Urban Road Standard NURS,2076 urban roads are further classified into
  - **Arterial roads:**
    - ➢ These roads are meant for thorough traffic on continuous route.
    - ➢ Parking, loading and unloading activities are generally restricted and regulated.
    - ➢ Pedestrians are allowed to cross only at the intersections or at designated crossing.
  - **Sub-arterial roads:**
    - ➢ These roads are somewhat lower level of travel mobility than arterial roads.
    - ➢ Parking, loading and unloading are restricted and regulated.
    - ➢ Pedestrian are allowed to cross only at the intersections or at designated crossing.
  - **Collector roads:**
    - ➢ It is intended for collecting and distributing the traffic to and from local roads and also for providing access to sub-arterial or arterial roads.
    - ➢ They may be located in industrial, residential or business areas.
  - **Local roads:**
    - ➢ It provided access to residence, business or other property.

- Such road doesn't carry large volume of traffic.

ii. **Technical/ Functional classification:**

- For assigning various geometric and technical parameters for design, roads are categorized into classes as follows:

Class I:
- ✓ They are highest standard roads with divided carriageway and access control with ADT 20,000 PCU or more in 20 years perspective period.
- ✓ Design speed adopted for design of this class roads in plain terrain is 120 kmph.

Class II:
- ✓ These are the roads with ADT of 5000-20000 PCU in 20 years perspective period.
- ✓ Design speed adopted for the design of this class of roads in plain terrain is 100 kmph.

Class III:
- ✓ They are the roads with ADT of 2000-5000 PCU in 20 years perspective period.
- ✓ Design speed adopted for design of this class of roads in plain terrain is 60 kmph.

Class IV:
- ✓ These are the roads with ADT of less than 2000 PCU in 20 years perspective period.
- ✓ Design speed adopted for design of this class of roads in plain terrain is 60 kmph.

2. **Calculate the minimum sight distance required to avoid head on collision of two cars approaching from the opposite direction, if both cars are at a speed of 90 km/hr. Assume a total perception and break reaction time of 2.5 seconds, coefficient of friction of 0.7 and a brake efficiency of fifty percent.**

Answer:

Given,

Speed of both cars, V=90 km/hr
Reaction time, t=2.5 s
Coefficient of friction, f=0.7
Brake efficiency = 50%
$g=9.81\ m/s^2$

For a head-on collision, the required sight distance is the sum of the stopping distances of both vehicles.

Due to 50% brake efficiency:

$f_e=0.5\times0.7=0.35$
V=90/3.6 =25 m/s

Stopping Sight Distance:

$$SSD= V_t+\frac{v^2}{2gfe}$$

$$= (25)\ (2.5) + \frac{25^2}{2(9.81)(0.35)}$$

$$=62.5+ \frac{625}{6.867}$$

$$=153.51\ m$$

Since both cars are approaching each other, both travel their stopping distances during the same period:

$$SSD = SD_1 + SD_2$$

Since both have the same speed:

$$SSD = 2(153.51)$$
$$SSD \approx 307.0 \text{ m}$$

3. **Explain why design, construction and maintenance of hill road of Nepal need special considerations? What are the special points to be considered in the alignment of hill road?**

Answer:

Hilly region of Nepal covers the mountainous & steep terrain having a cross slope of 25 to 60% and more. The road passing through the hilly terrain with a cross slope of 25% or more is generally termed as Hill Road. A hill road usually consists of either a river route or a ridge route. Road alignment in Nepal's hilly terrain is governed by a combination of geological, topographic, drainage, economic, and geometric design factors. The steep, unstable slopes and complex geology of the Himalayas and Middle Hills make the design and construction of hill roads more critical. The failure of hill roads may occur due to landslides and slope failure, efficient drainage problems, bridge failures etc. For example: BP highway constructed in the assistance of Japan has some commendable drainage arrangements but also some part of this road section suffer blocakges due to landslide and slope failure. Hence, design, construction and maintenance of hill road of Nepal need special considerations.

The special points to be considered in the alignment of hill road are given below:

i. **Geological Stability:**
   - Road alignment should pass through stable hill slopes and should avoid landslide and rockfall prone zones.
   - Construction activities might trigger landslide and rockfalls if the road alignment is to be passed through the area having unstable or weak hill slopes.
   - Excessive cutting works through hard solid rocks during construction must be avoided.

ii. **Cross drainage structures:**
   - Hilly region consists of large number of fast flowing, steep rivers that requires numerous cross-drainage works.
   - The road alignment should be selected such that the roadway consists of minimum number of cross drainage structures.

iii. **Geometric Design:**

- Geometric design of hill roads must be done according to the design guidelines and standards for hill region.
- The alignment should have minimum number of sharp curves and hairpin bends, gentler gradient.

iv. **Availability of construction materials:**

- The road alignment should be selected in such a way that the construction materials are available near the construction site.
- It reduces the transportation cost of materials and makes the construction more economical.

v. **Environment and weather:**

- Rainfall (or Snowfall) $\propto$ Altitude
- As the altitude decreases, the number of cross-drainage works required increases.
- At higher altitudes, the road pavement may witness snowfall during winter. This is why the alignment of hill roads should preferably be provided at an altitude between 900 m slopes exposed to sun.

vi. **Length:**

- The cost of construction of a hill road per kilometer length is comparatively very high.
- Hence, it should be ensured that the length of the road connecting two stations should be minimum possible.

4. **Define traffic engineering. A vehicle of weight 2 tonnes skids through a distance equal to 50 meter before colliding with another parked vehicle of weight 1 tonne. After collision, both the vehicles skid through a distance equal to 15 meters before stopping. If the weight of both vehicles are equal, compute the initial speed of moving vehicle. Take coefficient of friction as 0.4.**

Answer:

Traffic engineering is a branch of engineering that covers the road range of engineering applications with a focus on the safety of the public, the efficient use of transportation resources, and the mobility of people and goods. The objectives of traffic engineering are explained below:

i. To ensure road safety by reducing traffic accidents.
ii. To maintain efficient traffic movement by minimizing traffic congestion and improving traffic flow and travel speed.
iii. To increase the road capacity and maximizing the efficient use of existing road infrastructure.
iv. To reduce travel time and cost while shorten journey times.
v. To promote environmental sustainability by reducing air pollution, noise pollution and greenhouse gas emissions.

Given,

Weight of moving vehicle, W_1=2 tonnes
Weight of parked vehicle, W_2=1 tonne
Skid distance before collision, S_1=50m
Skid distance after collision, S_2=15m
Coefficient of friction, f=0.4
$g=9.81 m/s^2$

Let:

V_1 = initial speed before braking
V_2 = speed just before collision
V_3 = common speed of both vehicles immediately after collision

After collision, both vehicles skid together for 15m and stop.

Using,

$V_3^2=2gfS_2$
$V_3=\sqrt{2(9.81)(0.4)(15)}$
$V_3=10.85m/s$

Apply conservation of momentum:

$m_1V_2+m_2(0)=(m_1+m_2)V_3$

Since weight is proportional to mass,

$W_1V_2=(W_1+W_2)V_3$
$2V_2=(2+1)(10.85)$
$V_2=\frac{3(10.85)}{2}$
$V_2=16.28m/s$

The moving vehicle skids 50m before collision.

Using,

$V_1^2=V_2^2+2gfS_1$
$V_1^2=(16.28)^2+2(9.81)(0.4)(50)$
$V_1^2=265.04+392.4$
$V_1=\sqrt{657.44}$
$V_1=25.64m/s$

Convert into km/hr:

$V_1=25.64\times3.6$
$V_1\approx92.3$ km/hr

5. **a. Define tack coat, seal coat and priming.**
b. Discuss the types of seal coat and process of their application.
c. State in brief the classification of road maintenance activities in Nepal.

Answer:

a. Tack coat:

It is the single initial application of bituminous material on existing pavement surface which is relatively impervious such as existing bituminous, cement concrete or a pervious surface like WBM which has already been treated by the prime coat.

Seal coat:

It may be defined as the very thin surface treatment or single coat surface dressing which is either applied as a final step in the construction of certain bituminous surfaces to existing surfaces, which have cracked or worn out.

Prime coat:

It is the first application of low viscosity liquid bituminous material and is applied to the existing bases of the pervious surface like WBM road before the application of any bituminous treatment to the surface of the road.

b.

c. Highway Road maintenance is the task of preserving, repairing, and restoring a system of roadways to keep their condition as normal as possible and to ensure a long service life. It involves strategic planning, programming, and field implementation to provide better service facilities and a safer riding experience.

Based on the sources, the types and methods of road maintenance are categorized as follows:

The Department of Roads (DoR) in Nepal classifies maintenance activities into four primary categories:

i. **Routine Maintenance:** These are frequent, year-round activities performed on all elements of the highway to ensure serviceability in all weather conditions. Examples include grass and bush cutting, cleaning drains and culverts, and repairing minor structural damage.
ii. **Recurrent Maintenance:** These are localized activities carried out at regular intervals (typically six months to two years). They include sealing cracks, localized surface treatments, and repairing depressions and ruts.
iii. **Periodic Maintenance:** Conducted at intervals of several years, these activities improve the overall functionality of the road. Scope includes renewing the wearing course, restoring road markings, and repainting metal bridges.
iv. **Special Maintenance:** These are specialized or emergency works done to solve sudden problems like landslides or structural failures. Activities include removing debris, constructing temporary diversions, and strengthening pavement structures.

Maintenance can also be grouped based on its proactive or reactive nature:

i. **Preventive Maintenance:** Actions taken to prevent premature deterioration and retard the progression of deficiencies to increase the pavement's useful life.
ii. **Corrective Maintenance:** Actions taken to repair hazardous deficiencies and keep the highway within a tolerable level of serviceability.

6. **Describe the construction procedure for single or double bituminous surface dressing widely in Nepal by DoR. What do you know about cutback bitumen and bitumen emulsion?**

Answer:

Surface dressing refers to the thin surface coverings of bituminous layer and mineral aggregate. Bituminous surface dressing prevents the removal of binding materials from WBM road surface. Surface dressing may be classified into: single coat surface dressing and double coat surface dressing.

Single coat surface dressing:

It includes single application of thin layer of bitumen followed by the cover material of specified size of stone aggregate, which is then compacted by rollers.

Double coat surface dressing:

In this type of surface treatment, immediately after the laying of first coat, second application of binder is applied and followed by uniform spreading of the cover materials of smaller size aggregate and then rolled.

Construction procedure for single or double bituminous surface dressing:

i. Preparation of existing surface:
   - Surface is prepared to proper profile before treatment, with proper rectification of ruts and depression.
   - All the dust particles or loose materials are removed.
   - Prime coat is applied if the existing base course has pervious surface.

ii. Application of binder:
   - Uniform spraying of bituminous binder is done at the specified rate on a prepared surface using mechanical sprayer.
   - Excessive binder must not be applied at localized areas as it results in bleeding.

iii. Application of stone chipping:
   - Cover material is spread as per the requirement to cover the surface uniformly.

iv. Rolling of first coat:
   - Rolling is done with tandem roller (6 to 8 tonnes) starting from the edges proceeding towards the center longitudinally with overlapping not less than one third of the roller thread.

v. Application of binder and stone chipping for second coat:
   - Immediately after the application of binder to the prepared surface, cover materials is spread as before.
   - Rolling is done in the safe manner.

vi. Finishing and opening to traffic:
   - The surface is checked for longitudinal and cross profile using a straight edge of length 3 m and variation in surface greater than 6 mm are corrected.
   - The road section is opened to traffic after 24 hours.

Cutback bitumen:
Cutback bitumen is produced by dissolving or blending ordinary bitumen with a volatile petroleum solvent to reduce its viscosity and make it fluid enough for application at relatively low temperatures.
Common solvents: kerosene, naphtha and gasoline.

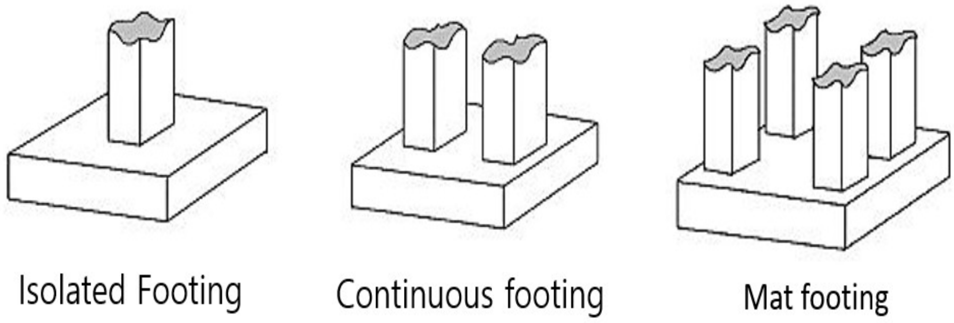
Bitumen emulsion:
Bitumen emulsion is a mixture of very small droplets of bitumen dispersed in water, with the help of an emulsifying agent. The bitumen droplets remain dispersed in water and can therefore be applied without heating the bitumen to high temperatures.
Types of bitumen emulsion:

i. **Cationic emulsion** – positively charged droplets
ii. **Anionic emulsion** – negatively charged droplets
iii. **Non-ionic emulsion** – no significant electrical charge

7. **a. What are the main types of foundation used for bridges?**
b. Describe in detail about underpinning with an aim to stabilize foundations of a bridge?
Answer:
a. In bridge construction, foundations are structural elements designed to receive loads from the superstructure and transfer them safely to stable soil or rock layers. These foundations are broadly classified into:
Shallow Foundations and Deep Foundations.

i. Shallow foundations:
   - These are used when firm soil or sound rock with adequate bearing capacity is available near the ground surface.
   - They are typically defined as foundations where the depth is less than or equal to the width.

- Suitable for smaller bridges, residential-scale crossings, or bridges in hilly regions where foundations can be placed directly on sound rock.
- Used when the soil with adequate bearing capacity is available near the ground surface.



ii. **Deep foundations:**
Deep foundations are required when the upper soil layers are weak, compressible, or susceptible to erosion (scour), necessitating the transfer of loads to stronger strata at greater depths.

- **Pile Foundations:**

  - ✓ Consists of long, slender members that are either driven or bored into the ground.
  - ✓ They transmit loads to deeper soil strata through end bearing (resting on a hard layer), skin friction (adhesion along the pile sides), or both.
  - ✓ Preferred for bridges when the bearing capacity of surface soil is low or for heavy structures in waterlogged areas.

- **Well Foundations (Caissons):**

  - ✓ These are large, watertight structures that are gradually sunk into the ground by excavating material from within them.
  - ✓ Load is transferred through base resistance and side resistance after the well reaches its required depth.
  - ✓ Specifically preferred for bridge piers in rivers as they provide strong resistance to heavy lateral forces and can be sunk below the scour depth to ensure stability during floods.

- **Pier Foundations:**

  - ✓ These are vertical columns similar to piles but typically larger in diameter, used to transfer structural loads to a hard stratum at depth.

The choice of foundation depends on several critical factors:

i. **Geological Conditions:**
- Foundations are placed on firm, non-erodible strata whenever possible to ensure reliability.

ii. **Scour Depth:**
- For river bridges, the foundation must be fixed below the calculated maximum scour depth to prevent the structure from being undermined during high-velocity flood events.

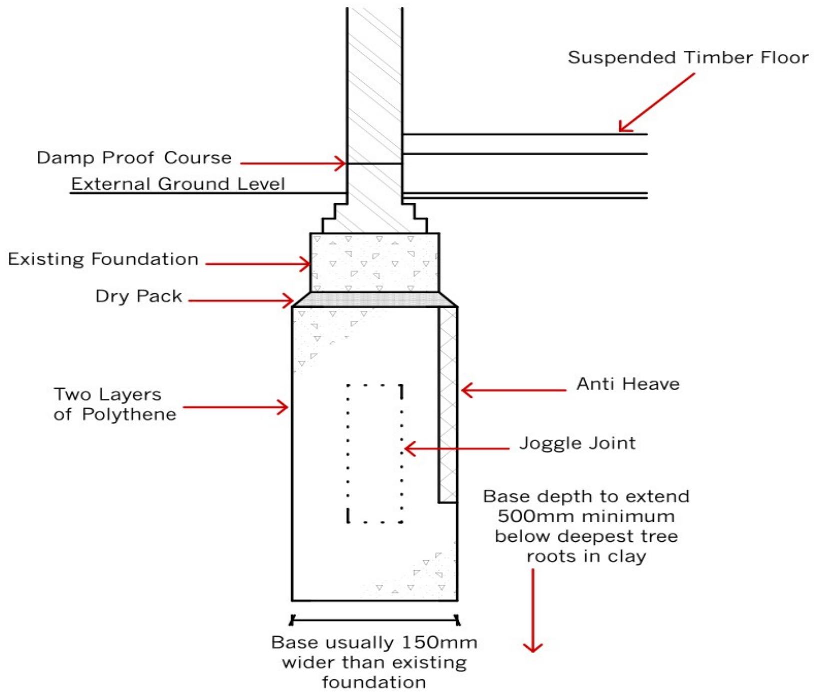
iii. **Bridge Type:**
- Heavier structures like arch bridges require very strong foundations, whereas lighter steel truss or cable-supported bridges may be feasible on slightly weaker strata.

b. The process of strengthening, supporting, or extending an existing bridge foundation so that it can safely carry the existing and additional loads and prevent further settlement, sliding, tilting, or scour-related instability is known as underpinning. It is adopted when the existing foundation becomes inadequate because of increased loads, foundation settlement, scour, deterioration, change in soil conditions, or construction of nearby structures. The main objectives are:

i. To increase the load-carrying capacity of the foundation.
ii. To stabilize a foundation affected by river scour or erosion.
iii. To improve the foundation's resistance to sliding and overturning.
iv. To extend the foundation to a deeper and stronger stratum.
v. To strengthen an old or deteriorated foundation.

Methods of underpinning:

i. Mass concrete underpinning:
   - The existing foundation is divided into small sections called bays.
   - Excavation is carried out below one bay at a time.
   - Concrete is placed beneath the existing foundation.
   - After concrete gains sufficient strength, work is carried out for another bay.



ii. Needle beam underpinning:
   - Needle beams are inserted through or below the existing foundation and supported by temporary props or new foundations.
   - The load from existing foundation is transferred to the needle beam and then to supporting elements.
   - New foundation members can be constructed subsequently.

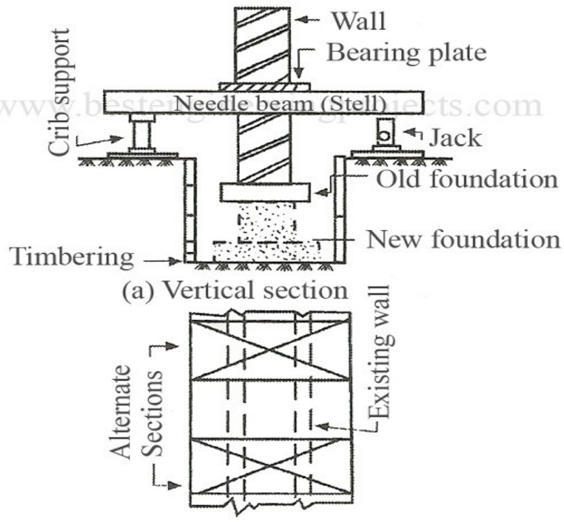
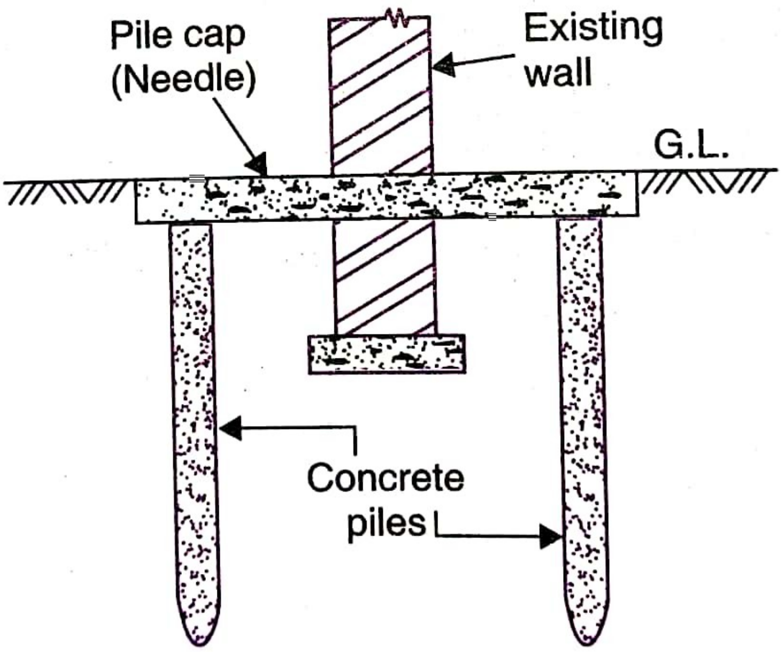


Fig 1 Underpinning by Pit Method

iii. Pile underpinning:
- Small-diameter piles are installed beside or through the existing foundation.
- Piles extend to a deeper, stronger stratum.
- A pile cap or reinforced concrete beam connects the piles to the existing foundation.
- Loads are transferred from the existing foundation to the piles.



8. **a. Suggest methods for improving the bearing capacity of weak soil for making foundations.**
b. Describe in brief the field methods of exploration of soil strata and survey for the construction of foundation of a bridge.

Answer:

a. Bearing capacity of soil is defined as maximum load per unit area that soil mass can safely withstand without undergoing shear failure or excessive settlement. It is the ability of a soil mass to safely support structural loads without any failure. The process of improvement of soil or strong foundation so that the soil can safely support the structural load without excessive settlement or shear failure is known as improvement of bearing capacity. Some methods for improving the bearing capacity of weak soil for making foundations are as follows:

i. Soil compaction:
- Soil compaction is the process of densification of soil by removing the air voids present in the soil particles under the action of mechanical energy.
- Compaction is measured in terms of dry density.
- It done by using rollers, vibratory compactors, rammers etc.

- It reduces the voids and increase soil density and shear strength.

ii. Preloading:

- Temporary load greater than or equal to the proposed foundation load is applied on the land.
- It increases the shear strength of soil and reduces future settlement.

iii. Stone columns:

- Stone columns consists of compacted gravel or crushed stone through weak soil.
- They increase bearing capacity of soil and also provides drainage and reinforcement.

iv. Sand compaction piles:

- Sand columns are installed and compacted in loose or soft soil.
- It increases the density and bearing capacity of the soil while reducing settlement.

v. Grouting:

- Chemical, cement or other suitable grout is injected into the soil.
- The void is filled and soil strength and stiffness is increased.
- It is useful where excavation is difficult or access is restricted.

vi. Chemical stabilization:

- Stabilizing agents such as cement, lime or fly ash is mixed with soil.
- It is effective for improving the bearing capacity of clayey soils.

vii. Deep foundations:

- If the soil is weak up to the considerable depth, we can use pile foundations or other deep foundations to transfer the structural loads to stronger soil strata at greater depth.

b.

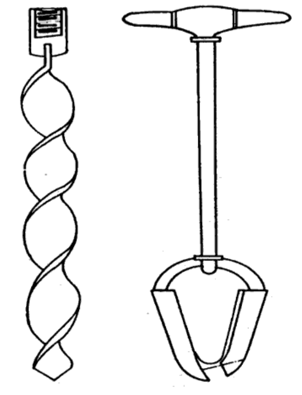
The field method of exploration of soil strata are listed below:

i. Trial pits:

- Excavation of pits at selected locations to expose the soil profile.
- Suitable for shallow exploration.
- Allows direct visual examination and collection of disturbed/undisturbed samples.
- Not suitable for very deep exploration or locations below the groundwater table.

ii. Auger boring:

- It is suitable for soft clay, silt, sandy clay and loam soil above the water table.
- A hand or mechanical auger is used to drill into the soil.
- Soil samples are obtained at different depths.



(a) HELICAL AUGER (b) POST-HOLE AUGER.

iii. Wash boring:

- Soil is loosened by water jet and the soil-water slurry is removed through a casing.
- Suitable for exploring deep soil deposits.
- Often used together with the Standard Penetration Test (SPT).

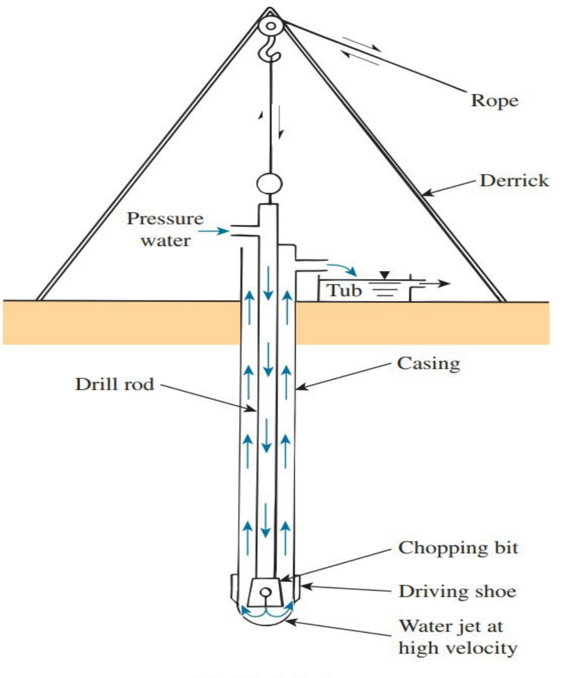
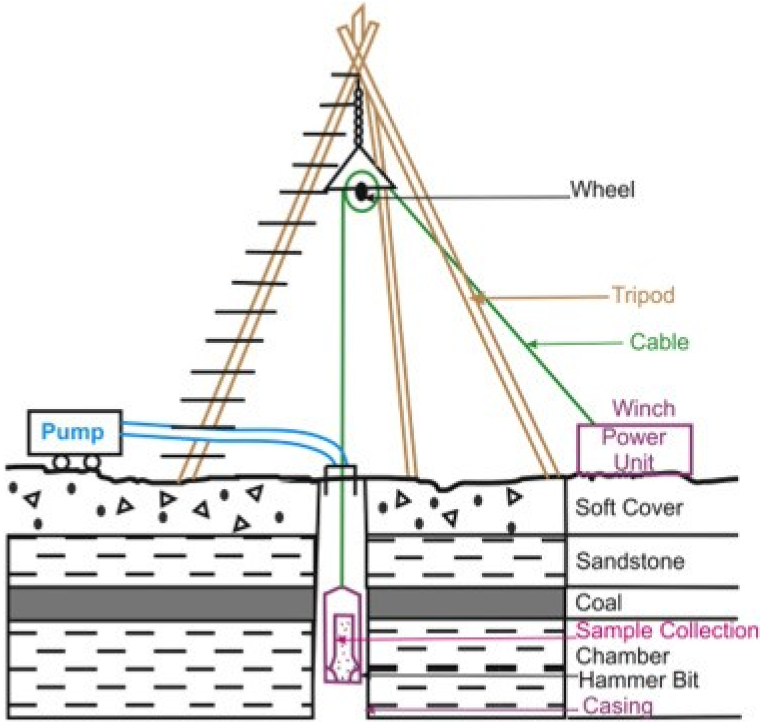


Fig. Wash Boring

iv. Percussion boring:

- Soil is broken by repeated blows of a heavy cutting tool and the material is removed.
- Useful in hard strata, gravel and boulder deposits where application of auger is difficult.



v. Rotary drilling:
- ➢ A rotating drill bit is used to penetrate soil and rock.
- ➢ Provides continuous or selected samples/core specimens.
- ➢ Particularly useful for deep bridge foundations and rock exploration.

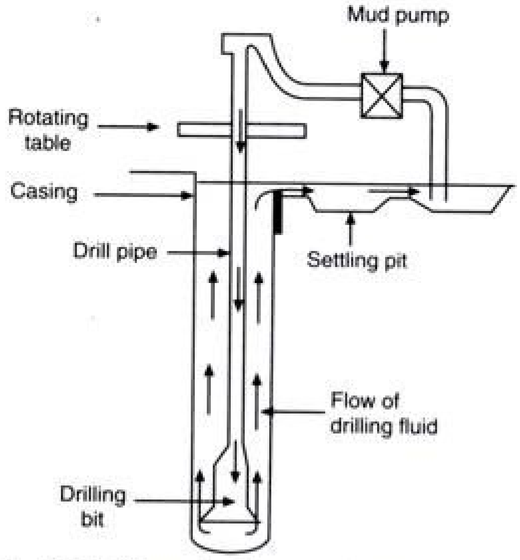
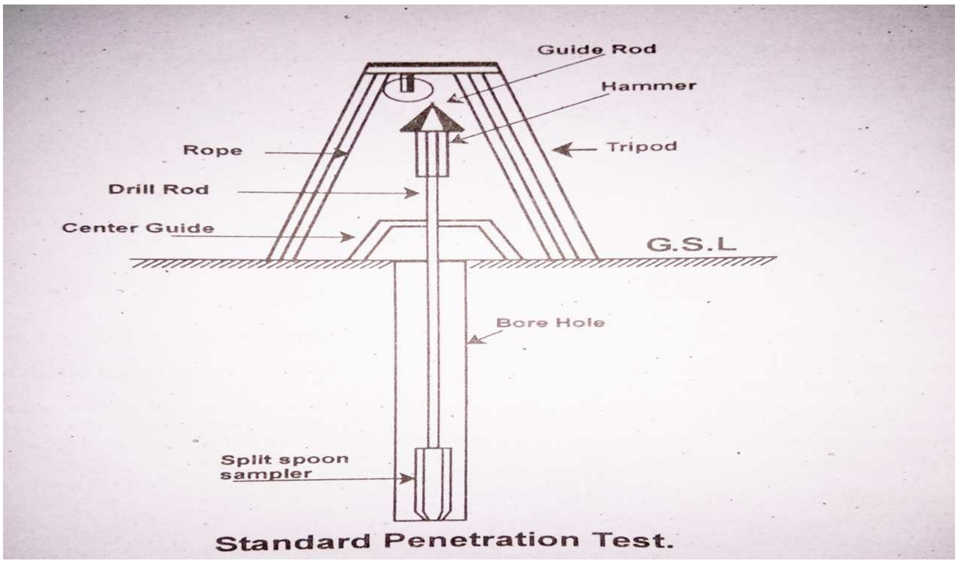


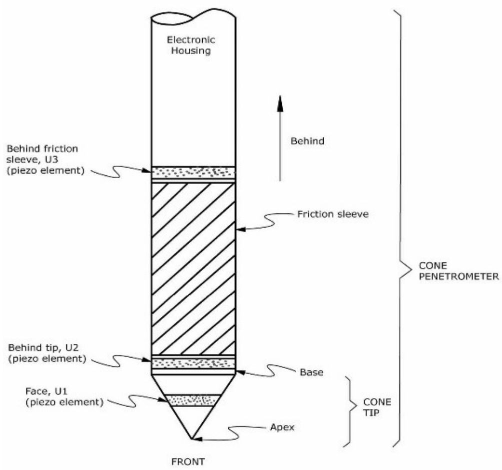
Fig. 18.8. Schematic diagram of rotary method

vi. Standard Penetration Test:
- ➢ SPT tests is one of the most commonly used field tests.
- ➢ The test is conducted in a borehole using a split-spoon sampler from where N-value is determined.
- ➢ The number of blows required to drive the sampler 300 mm is called SPT N-value.



vii. Cone Penetration Test:

- In cone penetration test, a cone is pushed into ground at a constant rate of 2 cm per second.
- It measures cone resistance and sleeve friction.
- The number of blows of hammer required to penetrate a cone to 300 mm depth is called CPT N_c value.



The methods of survey for the construction of foundation of bridge:

i. **Topographical Survey**
The topographical survey provides the physical data of the terrain to fix the bridge alignment and determine foundation locations.

a. **Contour Mapping:** Used to understand the ground levels, identify stable banks, and determine the storage or flow characteristics of the area.
b. **Instruments:** Modern surveys utilize total Stations for precise angular and distance measurements, and GNSS/GPS for accurate geographic coordinates.
c. **Cross-Section and Longitudinal Sectioning:** Detailed sections of the riverbed and banks are surveyed to plan the bridge span and foundation levels.

ii. **Hydrological Survey**
Hydrological surveys are critical for determining the depth of the foundation to prevent failure due to water-induced erosion (scour).

a. **Discharge Measurement:**
   - **Current Meter Method:** The stream cross-section is divided into vertical strips, and a current meter measures velocity to compute the total discharge.
   - **Slope-Area Method:** An indirect technique using high-water marks and Manning's equation to estimate peak flood discharge.

b. **Scour Investigation: Lacey's formula** is commonly used to calculate normal and maximum scour depth. Foundations must be fixed below the calculated maximum scour level to ensure stability during floods.

iii. **Geotechnical Investigation (Soil Strata Exploration)**
This is the process of acquiring information about the soil and rock layers beneath the ground surface to determine their bearing capacity and engineering properties.

a. **Boring Methods:**
- **Auger Boring:** Suitable for soft clay, silt, and sandy soil, generally for shallow exploration above the water table.
- **Wash Boring:** Used for loose sandy or clayey soils below the water table.
- **Rotary Drilling:** Best for deep tube wells and exploring medium to hard formations.
- **Percussion Boring:** Employed for hard clay, gravel, boulders, and weathered rock.

b. **In-situ (Field) Testing:**
- **Standard Penetration Test (SPT):** Conducted within a borehole using a split-spoon sampler. The **N-value** (blows per 300 mm) is used to calculate the bearing capacity for footings, piles, and wells.
- **Cone Penetration Test (CPT):** A cone is pushed into the ground at a constant rate to measure resistance and friction without a borehole.
- **Dynamic Cone Penetration Test (DCPT):** Useful for rapid assessment in hilly areas with boulders and gravel where boreholes are difficult to drill.
- **Plate Load Test:** Measures actual settlement under incremental loads at the foundation level.

c. **Groundwater Table (GWT) Observation:** Monitoring the position of the GWT is essential, as its rise can significantly reduce the soil's bearing capacity and increase hydrostatic pressure.

iv. **Laboratory Testing**
Soil and rock samples collected during boring are analyzed in a laboratory to determine index properties (classification) and engineering properties such as cohesion (c), angle of internal friction (φ), and unit weight. These values are then used in theoretical equations, like Terzaghi's theory, to confirm the safe bearing capacity.

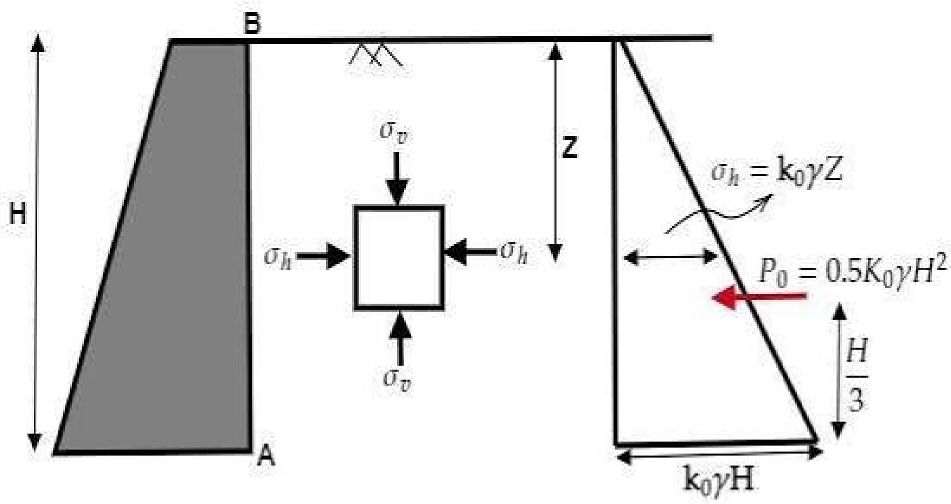
9. **Mention about Rankine's earth pressure theory.**

Answer:

The expression for active and passive earth pressure on the vertical back face of earthwork supporting granular soil having horizontal surface is given by Rankine's theory.

The assumptions for Rankine's theory:

i. The back of the retaining wall is vertical and smooth (no wall friction).
ii. The backfill is dry, homogeneous, cohesionless (granular) soil.
iii. The ground surface is horizontal.
iv. The soil is in the plastic equilibrium state.
v. Failure occurs along a plane surface.



Let us consider a soil element at a depth h below the ground surface.

$$Vertical\ \ stress\ \ due\ \ ¿\ \ overburden:$$

$$\sigma_v = \gamma h$$
$$where:$$

$$\gamma = unit\ \ weight\ \ of\ \ soil\left(kN/m^3\right)$$

$$h = depth\ \ below\ \ ground\ \ level\left(m\right)$$

$$The\ \ horizontal\ \ stress\ \ under\ \ active\ \ condition\ \ is$$

$$\sigma_h = K_a \sigma_v$$

$$where\ \ K_a\ \ is\ \ the\ \ coefficient\ \ of\ \ active\ \ earth\ \ pressure.$$

$$¿\ \ Mohr's\ \ \circ\wedge\ \ Rankine's\ \ failure\ \ criterion,$$

$$K_a = \frac{1 - \sin\phi}{1 + \sin\phi}$$

or

$$K_a = \tan^2\left(45^\circ - \frac{\phi}{2}\right)$$

$$where$$

$$\phi\ \ =\ \ angle\ \ of\ \ internal\ \ friction\ \ of\ \ soil.$$

$$Substituting$$

$$\sigma_h = K_a \gamma h$$

$$For\ \ a\ \ retaining\ \ wall\ \ of\ \ height\ \ H,$$

$$Pressure\ \ at\ \ the\ \ top:$$

$$p = 0$$

$$Pressure\ \ at\ \ the\ \ bottom:$$

$$p = K_a \gamma H$$

The total active earth pressure equals the area of the pressure triangle.

$$P_a = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$P_a = \frac{1}{2}(K_a \gamma H)(H)$$

Therefore,

$$P_a = \frac{1}{2} K_a \gamma H^2$$

This force acts horizontally on the wall.

Since the pressure distribution is triangular, the resultant acts at the centroid of the triangle.

Hence,

$$\text{Point of application} = \frac{H}{3} \text{ above the base}$$

At passive failure, Rankine's passive earth-pressure coefficient is:

$$K_p = \frac{1 + \sin\phi}{1 - \sin\phi}$$

¿,

$$K_a = \tan^2\left(45^\circ + \frac{\phi}{2}\right)$$

The lateral passive pressure at depth z is:

$$\sigma_{hp} = K_p \gamma z$$

At the bottom of the wall:

$$\sigma_{hp,max} = K_p \gamma H$$

The total passive earth pressure is therefore:

$$P_p = \frac{1}{2} \gamma H^2 \tan^2\left(45^\circ + \frac{\phi}{2}\right)$$

Hence,

The resultant P_p *also acts at* $\frac{H}{3}$ above the base.

10. **Explain the well sinking operation during bridge construction. How to avoid 'tilt and shifts 'and measures to correct "tilts and shifts".**

Answer:

Well or caisson foundation is the watertight structure used in bridge pier, dam construction etc. It is a readymade hollow cylinder depressed into the soil up to the desired level and then filled with concrete that converts into foundations.
Well sinking operation during the bridge construction is given in sequence below:

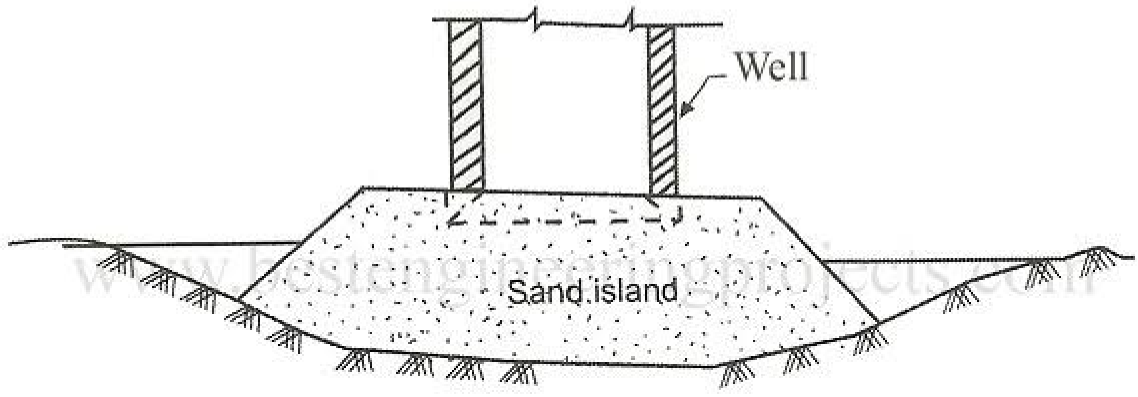


Fig 1 A Typical Sand Island

i. Preparation of well curb:

- The well is constructed with a **cutting edge/well curb** at its bottom.
- The cutting edge is generally made of steel and reinforced concrete.
- It facilitates penetration of the well into the river bed.

ii. Construction of well steining:

- The well steining is constructed over the curb in stages.
- The steining provides the required strength and weight for sinking.
- The well is usually constructed above the river bed or water level using suitable staging.

iii. Excavation of well:

- Soil is excavated from inside the well using chisels, chamshells. Grab buckets and so on.
- As soil is removed from inside, the well gradually sinks under its own weight.

iv. Addition of steining:

- As the well sinks, additional sections of steining are constructed at the top.

v. Control of tilt and shift:

- During sinking, the **verticality and horizontal position** of the well are continuously checked using surveying instruments and plumb lines.

vi. Foundation:
When the well reaches the required depth:

- Excavation is stopped.
- The bottom is cleaned.
- The bottom plug is constructed, generally using concrete.
- The well may then be filled with sand or other approved material.

- A top plug and **well cap** are constructed.
- The bridge pier is finally constructed over the well cap.

The inclination of the well from its intended vertical position is called tilt. The horizontal displacement of the well from its designed position is called shift.

Some of the methods to avoid excavation:

i. Soil must be excavated uniformly from all sides of well.
ii. Excessive excavation on one side must be avoided.
iii. Maintain the verticality of well by checking frequently with plumb bob, and other surveying instruments.
iv. Dredging operation must be carried out carefully.
v. Kentledge load should be applied symmetrically and carefully.
vi. Steining should be constructed uniformly so that well has adequate stiffness and weight.

Some measures to correct tilt and shifts are given below:

i. Eccentric dredging can be carried out by excavating more soil on the lower side of the well.
ii. Eccentric kentledge load may be applied on the higher side of tilted well.
iii. The well may be pulled by using ropes, hydraulic jacks etc.
iv. Water jetting can be used to cut the edge to reduce soil resistance on the appropriate side.
v. Use of temporary struts or structural members may be used in appropriate situations to control or correct the position.